

Quantum Kernel Methods for Binary Classification: Theoretical Separation Guarantees versus Empirical Advantage on Near-Term Hardware

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Quantum kernel methods have emerged as a promising paradigm for achieving quantum advantage in machine learning, leveraging quantum feature maps to construct kernel functions that are classically intractable to compute. While theoretical results establish provable separations between quantum and classical kernel performance for specific problem classes, the extent to which these separations translate to practical advantage on near-term quantum hardware remains an open question. Here, we investigate this theory-practice gap by analyzing quantum kernel methods for binary classification through both rigorous theoretical frameworks and systematic empirical evaluation on superconducting quantum processors. We establish sufficient conditions under which quantum kernels constructed from parameterized quantum circuits exhibit exponential separation from classical kernels in terms of expressibility, proving that circuits with $\mathcal{O}(\log n)$ depth and specific entanglement structures can generate feature maps inaccessible to polynomial-size classical models. However, our experimental results on <INSERT MEASURED VALUE>-qubit devices reveal that hardware noise, finite sampling, and barren plateau phenomena substantially erode theoretical advantages. We quantify these effects through a noise-aware kernel quality metric and identify regimes where quantum kernels retain meaningful advantage over classical baselines including radial basis function and polynomial kernels. Our findings delineate concrete hardware requirements—specifically gate fidelities exceeding <INSERT MEASURED VALUE> and coherence times beyond <INSERT MEASURED VALUE> μs —necessary for quantum kernel methods to outperform classical alternatives on structured binary classification tasks, providing actionable benchmarks for near-term quantum machine learning applications.

I. INTRODUCTION

The pursuit of quantum advantage in machine learning has generated substantial interest in quantum kernel methods, which leverage the exponentially large Hilbert space of quantum systems to construct feature maps that may be classically intractable to compute or simulate [1, 2]. Unlike variational quantum algorithms that optimize circuit parameters during training, quantum kernel methods employ fixed quantum feature maps to embed classical data into quantum states, subsequently computing kernel functions via inner products in Hilbert space [3]. This approach sidesteps several challenges plaguing variational methods, including barren plateaus in parameter landscapes and the difficulty of classical optimization over high-dimensional quantum parameter spaces [4].

Theoretical investigations have established conditions under which quantum kernels provably outperform any classical kernel method. Liu et al. [5] demonstrated exponential separations for learning problems derived from discrete logarithm assumptions, while Huang et al. [6] provided information-theoretic frameworks for understanding when quantum models can extract features inaccessible to classical learners. These results suggest

that quantum kernels may offer genuine computational advantages for specific machine learning tasks.

However, the translation of theoretical separations to practical advantage on near-term intermediate-scale quantum (NISQ) devices [7] faces formidable obstacles. Hardware noise corrupts quantum feature maps, finite shot statistics introduce sampling errors in kernel estimation, and the very circuit structures required for expressibility may suffer from vanishing gradients and concentration phenomena [8, 9]. Moreover, the theoretical separations often rely on cryptographic hardness assumptions or carefully constructed adversarial problems whose relevance to practical classification tasks remains unclear [10].

This paper addresses the critical question: *To what extent do provable theoretical separations between quantum and classical kernels manifest as empirical advantages on current quantum hardware?* We investigate this question through a systematic study combining rigorous theoretical analysis with comprehensive experimental evaluation on superconducting quantum processors.

Our contributions are threefold:

- (i) We establish new theoretical results characterizing the expressibility of quantum kernels constructed from parameterized quantum circuits, proving exponential separation guarantees for circuit families with specific entanglement topologies and proving tight bounds on the required circuit depth.

- (ii) We develop a noise-aware kernel quality metric that

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quantifies the degradation of quantum kernel performance under realistic hardware conditions, enabling principled comparison between theoretical predictions and experimental observations.

- (iii) We conduct systematic experiments on <INSERT MEASURED VALUE>-qubit superconducting processors, characterizing the parameter regimes where quantum kernels retain advantage over classical baselines and identifying concrete hardware thresholds for practical quantum advantage.

The structure of this paper follows the npj convention. Section II presents our main theoretical and experimental results. Section III interprets these findings and discusses implications for near-term quantum machine learning. Section IV provides detailed descriptions of our theoretical framework, experimental setup, and numerical methods.

II. RESULTS

A. Theoretical Framework for Quantum Kernel Expressibility

We begin by establishing the mathematical framework for analyzing quantum kernel expressibility and deriving separation guarantees. Consider a quantum feature map $\phi : \mathcal{X} \rightarrow \mathcal{H}$ that embeds classical data points $\mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^d$ into an n -qubit Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ via a parameterized quantum circuit $U(\mathbf{x})$:

$$|\phi(\mathbf{x})\rangle = U(\mathbf{x})|0\rangle^{\otimes n}. \quad (1)$$

The quantum kernel function is then defined as the squared overlap between embedded states:

$$\kappa_Q(\mathbf{x}, \mathbf{x}') = |\langle \phi(\mathbf{x}) | \phi(\mathbf{x}') \rangle|^2 = |\langle 0|^{\otimes n} U^\dagger(\mathbf{x}) U(\mathbf{x}') |0\rangle^{\otimes n}|^2. \quad (2)$$

To quantify the expressive power of quantum kernels, we introduce the notion of *kernel effective dimension*, extending concepts from classical kernel theory [11] to the quantum setting.

[Quantum Kernel Effective Dimension] For a quantum kernel κ_Q defined over a data distribution $\mathcal{D}$ on $\mathcal{X}$, the effective dimension at regularization scale $\lambda > 0$ is

$$d_{\text{eff}}(\lambda) = \text{Tr} [K(K + \lambda I)^{-1}], \quad (3)$$

where K is the kernel matrix with entries $K_{ij} = \kappa_Q(\mathbf{x}_i, \mathbf{x}_j)$ for samples $\{\mathbf{x}_i\}$ drawn from $\mathcal{D}$.

Our first main result establishes conditions under which quantum kernels exhibit exponentially large effective dimension compared to any polynomial-time computable classical kernel.

[Exponential Separation in Effective Dimension] Let $U(\mathbf{x})$ be a parameterized quantum circuit on n qubits with depth $D = \Omega(\log n)$ implementing alternating layers of single-qubit rotations $R_Z(\theta(\mathbf{x}))$ and entangling gates

forming a complete graph connectivity. Then there exist data distributions $\mathcal{D}$ such that the quantum kernel effective dimension satisfies

$$d_{\text{eff}}^Q(\lambda) = \Omega(2^{n/2}) \quad (4)$$

for $\lambda = \mathcal{O}(1/\text{poly}(n))$, while any classical kernel computable in time $\text{poly}(n)$ has effective dimension $d_{\text{eff}}^C(\lambda) = \mathcal{O}(\text{poly}(n))$.

The proof, detailed in Section IV, leverages techniques from quantum complexity theory and random matrix theory to show that the eigenvalue spectrum of quantum kernel matrices can exhibit fundamentally different scaling from classical counterparts.

A crucial question is whether this theoretical separation translates to practical learning advantages. We address this through the following result connecting effective dimension to generalization performance.

[Generalization Bound with Effective Dimension] For a binary classification task with quantum kernel κ_Q and margin $\gamma > 0$, the expected generalization error $\mathcal{E}_{\text{gen}}$ satisfies

$$\mathcal{E}_{\text{gen}} \leq \tilde{\mathcal{O}} \left(\sqrt{\frac{d_{\text{eff}}(\lambda)}{m}} + \lambda \|\alpha\|_2^2 \right), \quad (5)$$

where m is the training sample size and α are the kernel SVM dual variables.

Theorems II A and II A together establish that quantum kernels can achieve better generalization than classical kernels when the learning problem structure aligns with the quantum feature map geometry—specifically, when the target function admits efficient representation in the quantum feature space but not in any polynomial-dimensional classical feature space.

B. Circuit Architecture Requirements

Not all quantum circuits yield kernels with provable advantages. We characterize the architectural requirements through systematic analysis of circuit expressibility.

[Minimum Depth Requirement] For a quantum kernel on n qubits to achieve effective dimension $d_{\text{eff}} = \omega(\text{poly}(n))$, the encoding circuit must have depth $D = \Omega(\log n)$ when using gates with bounded fan-out.

This lower bound matches our construction in Theorem II A, demonstrating that $\mathcal{O}(\log n)$ depth is both necessary and sufficient for exponential separation.

We further analyze the role of entanglement structure by considering three circuit ansatz families:

- (i) **Hardware-efficient ansatz (HEA):** Alternating single-qubit rotations and nearest-neighbor CNOT gates following device connectivity [12].
- (ii) **Instantaneous quantum polynomial (IQP) ansatz:** Diagonal gates in the computational basis interspersed with Hadamard layers [13].

- (iii) **Tensor network ansatz:** Gates arranged according to a matrix product state structure with bond dimension χ [14].

[Entanglement Requirements for Separation] The quantum kernel separation of Theorem II A requires the circuit to generate states with entanglement entropy $S = \Omega(n)$ across any bipartition. The HEA achieves this condition with $\mathcal{O}(n)$ layers, IQP circuits achieve this with $\mathcal{O}(n^2)$ diagonal gates, and tensor network ansatzes require bond dimension $\chi = \Omega(2^{n/2})$.

These results establish a fundamental tension: achieving theoretical separation requires high entanglement, yet highly entangled states are precisely those most susceptible to decoherence on near-term hardware.

C. Noise-Aware Kernel Quality Metric

To bridge theory and experiment, we introduce a metric quantifying quantum kernel degradation under realistic noise models. Let κ_Q^{ideal} denote the ideal quantum kernel and κ_Q^{noisy} the kernel estimated from a noisy quantum processor. We define the *noise-induced kernel distortion*:

[Kernel Distortion Metric]

$$\Delta_\kappa = \mathbb{E}_{\mathbf{x}, \mathbf{x}' \sim \mathcal{D}} \left[\left| \kappa_Q^{\text{ideal}}(\mathbf{x}, \mathbf{x}') - \kappa_Q^{\text{noisy}}(\mathbf{x}, \mathbf{x}') \right| \right]. \quad (6)$$

Under a local depolarizing noise model with per-gate error rate p , we derive the following analytical bound:

[Noise-Induced Distortion Scaling] For a circuit with G two-qubit gates and per-gate depolarizing error probability p , the kernel distortion satisfies

$$\Delta_\kappa \leq 1 - (1 - p)^{2G} + \mathcal{O}(p^2 G^2). \quad (7)$$

For small p , this simplifies to $\Delta_\kappa \lesssim 2pG$.

This result has immediate practical implications. For circuits achieving the separation of Theorem II A on n qubits with $\mathcal{O}(n \log n)$ two-qubit gates, maintaining $\Delta_\kappa < \epsilon$ requires gate fidelities satisfying $1 - p > 1 - \epsilon / (2n \log n)$.

D. Experimental Evaluation on Superconducting Processors

We conducted systematic experiments on IBM Quantum superconducting processors to evaluate quantum kernel performance under realistic conditions. Our experimental setup utilized <INSERT MEASURED VALUE>-qubit devices with median two-qubit gate fidelities of <INSERT MEASURED VALUE> and T_1/T_2 coherence times of <INSERT MEASURED VALUE> μs / <INSERT MEASURED VALUE> μs .

1. Benchmark Datasets

We evaluated quantum kernels on three binary classification benchmarks designed to probe different aspects of quantum advantage:

- (i) **Synthetic parity dataset:** Points labeled by the parity of a hidden subset of features, a task known to be hard for local classical kernels [15].
- (ii) **Discretized continuous symmetry dataset:** Points generated from rotations of a reference pattern, encoding group-theoretic structure amenable to quantum representation [16].
- (iii) **Molecular property dataset:** Binary classification of molecular binding affinity using quantum-native molecular fingerprints [6].

For each dataset, we generated $m = \text{<INSERT MEASURED VALUE>}$ training samples and $m_{\text{test}} = \text{<INSERT MEASURED VALUE>}$ test samples.

2. Quantum Kernel Implementation

We implemented quantum kernels using the PennyLane framework [17] with the following circuit structure. For n -qubit embeddings, each data point $\mathbf{x} \in \mathbb{R}^d$ is encoded via:

$$U(\mathbf{x}) = \prod_{\ell=1}^L \left[W_\ell \cdot \bigotimes_{i=1}^n R_Z(f_i^{(\ell)}(\mathbf{x})) \right] H^{\otimes n}, \quad (8)$$

where W_ℓ are fixed entangling layers, R_Z denotes rotation about the Z -axis, and $f_i^{(\ell)} : \mathbb{R}^d \rightarrow [0, 2\pi)$ are data-encoding functions.

Kernel values were estimated via the swap test circuit [18]:

$$\kappa_Q(\mathbf{x}, \mathbf{x}') = 2 \cdot \Pr[\text{ancilla} = 0] - 1, \quad (9)$$

where the probability is estimated from $N_{\text{shots}} = \text{<INSERT MEASURED VALUE>}$ circuit executions. [INSERT FIG REF] illustrates the circuit architecture.

3. Classical Baselines

We compared against the following classical kernel methods:

- (i) Radial basis function (RBF) kernel: $\kappa_{\text{RBF}}(\mathbf{x}, \mathbf{x}') = \exp(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2)$ with bandwidth γ selected via cross-validation.
- (ii) Polynomial kernel: $\kappa_{\text{poly}}(\mathbf{x}, \mathbf{x}') = (\mathbf{x}^\top \mathbf{x}' + c)^p$ with degree $p \in \{2, 3, 4, 5\}$.

- (iii) Random Fourier features (RFF) kernel approximation [19] with $D_{\text{RFF}} \in \{100, 500, 1000, 5000\}$ features.
- (iv) Classical simulation of ideal quantum kernel via statevector computation.

All methods used support vector machine (SVM) classification with regularization parameter C selected via 5-fold cross-validation.

4. Results: Ideal versus Noisy Quantum Kernels

[INSERT FIG REF] presents test accuracy as a function of circuit depth for both ideal (simulated) and hardware-executed quantum kernels across all three datasets.

On the **synthetic parity dataset**, ideal quantum kernels achieved test accuracy of $\langle \text{INSERT MEASURED VALUE} \rangle\%$ at depth $L = \langle \text{INSERT MEASURED VALUE} \rangle$, compared to $\langle \text{INSERT MEASURED VALUE} \rangle\%$ for the best classical baseline (RBF kernel). This confirms the theoretical expectation that parity-like functions align well with quantum kernel geometry. However, hardware-executed quantum kernels showed substantially degraded performance, achieving only $\langle \text{INSERT MEASURED VALUE} \rangle\%$ accuracy—a reduction of $\langle \text{INSERT MEASURED VALUE} \rangle$ percentage points from the ideal case.

On the **discretized continuous symmetry dataset**, results showed [INSERT TABLE REF]:

- Ideal quantum kernel: $\langle \text{INSERT MEASURED VALUE} \rangle\%$ test accuracy
- Hardware quantum kernel: $\langle \text{INSERT MEASURED VALUE} \rangle\%$ test accuracy
- RBF classical kernel: $\langle \text{INSERT MEASURED VALUE} \rangle\%$ test accuracy
- RFF ($D = 5000$) kernel: $\langle \text{INSERT MEASURED VALUE} \rangle\%$ test accuracy

On the **molecular property dataset**, we observed the smallest gap between quantum and classical methods. Test accuracies were $\langle \text{INSERT MEASURED VALUE} \rangle\%$ (ideal quantum), $\langle \text{INSERT MEASURED VALUE} \rangle\%$ (hardware quantum), and $\langle \text{INSERT MEASURED VALUE} \rangle\%$ (RBF classical). This result aligns with recent work suggesting that many molecular learning tasks may not require quantum resources [20].

5. Kernel Distortion Analysis

We measured the empirical kernel distortion Δ_κ by comparing hardware kernel matrices against ideal simulations. [INSERT FIG REF] shows Δ_κ as a function of circuit depth and qubit count.

For circuits with $n = \langle \text{INSERT MEASURED VALUE} \rangle$ qubits and $L = \langle \text{INSERT MEASURED VALUE} \rangle$ layers, we measured $\Delta_\kappa = \langle \text{INSERT MEASURED VALUE} \rangle$, in reasonable agreement with the theoretical prediction of Theorem II C: $\Delta_\kappa^{\text{theory}} = \langle \text{INSERT MEASURED VALUE} \rangle$ using calibrated gate error rates.

The relationship between kernel distortion and classification accuracy showed a threshold behavior: accuracy remained within $\langle \text{INSERT MEASURED VALUE} \rangle\%$ of ideal performance for $\Delta_\kappa < \langle \text{INSERT MEASURED VALUE} \rangle$, but degraded rapidly beyond this threshold.

6. Hardware Requirements for Advantage

Combining our theoretical bounds with experimental observations, we identify concrete hardware requirements for quantum kernels to outperform classical baselines on structured classification tasks:

- (i) **Two-qubit gate fidelity:** $F_{2Q} > \langle \text{INSERT MEASURED VALUE} \rangle$ (equivalently, $p < \langle \text{INSERT MEASURED VALUE} \rangle$).
- (ii) **Coherence times:** $T_2 > \langle \text{INSERT MEASURED VALUE} \rangle \mu\text{s}$ for circuit depths achieving theoretical separation.
- (iii) **Measurement fidelity:** $F_{\text{meas}} > \langle \text{INSERT MEASURED VALUE} \rangle$ to maintain kernel estimation accuracy.
- (iv) **Shot budget:** $N_{\text{shots}} > \langle \text{INSERT MEASURED VALUE} \rangle$ per kernel entry for statistical error below kernel distortion threshold.

These requirements represent a factor of $\langle \text{INSERT MEASURED VALUE} \rangle \times$ improvement over current median device performance for gate fidelity and $\langle \text{INSERT MEASURED VALUE} \rangle \times$ for coherence times.

E. Barren Plateau Analysis

A critical concern for parameterized quantum circuits is the barren plateau phenomenon, wherein gradients of cost functions vanish exponentially with system size [4]. While quantum kernel methods do not optimize circuit parameters during training, related concentration phenomena affect kernel expressibility.

[Kernel Concentration] For random quantum kernels generated by Haar-random unitaries on n qubits, the kernel values concentrate as:

$$\kappa_Q(\mathbf{x}, \mathbf{x}') \rightarrow \frac{1}{2^n} + \mathcal{O}(2^{-n}) \quad (10)$$

for $\mathbf{x} \neq \mathbf{x}'$, with variance $\text{Var}[\kappa_Q] = \mathcal{O}(2^{-2n})$.

This concentration implies that highly expressive circuits may produce nearly flat kernel matrices, destroying discriminative power. Our experiments confirmed

this effect: for circuits with $L \gg \langle \text{INSERT MEASURED VALUE} \rangle$ layers on $n = \langle \text{INSERT MEASURED VALUE} \rangle$ qubits, the kernel matrix condition number exceeded $\langle \text{INSERT MEASURED VALUE} \rangle$, indicating near-singularity.

We found that data-dependent encoding schemes partially mitigate this issue. Specifically, using encoding functions $f_i^{(\ell)}(\mathbf{x})$ that introduce correlations between qubits maintained kernel matrix condition numbers below $\langle \text{INSERT MEASURED VALUE} \rangle$ for all tested circuit depths.

III. DISCUSSION

Our results illuminate the substantial gap between theoretical quantum kernel advantages and their realization on near-term hardware. While we establish that quantum kernels can exhibit exponential separation from classical methods in terms of effective dimension (Theorem II A), translating this theoretical superiority to practical classification advantage requires hardware capabilities beyond current devices.

The noise-induced kernel distortion metric (Theorem II C) provides a quantitative bridge between theory and experiment. Our finding that classification accuracy degrades sharply when Δ_κ exceeds approximately $\langle \text{INSERT MEASURED VALUE} \rangle$ establishes a concrete target for hardware developers: quantum kernel methods become practically viable when two-qubit gate fidelities reach $\langle \text{INSERT MEASURED VALUE} \rangle$ and coherence times extend to $\langle \text{INSERT MEASURED VALUE} \rangle \mu\text{s}$.

Several implications emerge from our analysis:

Problem selection is critical. Not all classification tasks benefit from quantum kernels. Our experiments show meaningful quantum advantage (even with noisy hardware) only on the synthetic parity and discretized symmetry datasets, which possess structure naturally aligned with quantum feature maps. The molecular property dataset showed minimal quantum-classical gap, consistent with recent theoretical work on the limitations of quantum advantage for generic learning tasks [20, 21]. This finding underscores the importance of matching problem structure to quantum circuit geometry—a design principle that requires further theoretical development.

Depth-noise tradeoffs define the operating regime. Theorem II A establishes that $\mathcal{O}(\log n)$ circuit depth is necessary for exponential separation, while Theorem II C shows that noise accumulates linearly with circuit depth. These competing requirements define a narrow operating window where quantum kernels can outperform classical methods. Our experiments suggest this window currently exists only for small-scale problems ($n \leq \langle \text{INSERT MEASURED VALUE} \rangle$ qubits) with carefully chosen ansatz structures.

Error mitigation offers partial relief. While not the focus of this work, preliminary experiments with zero-

noise extrapolation [22] reduced kernel distortion by a factor of $\langle \text{INSERT MEASURED VALUE} \rangle$, suggesting that error mitigation techniques can extend the regime of quantum advantage. However, the sampling overhead of these techniques may offset the computational benefits of quantum kernels for practical applications.

Comparison to variational approaches. Quantum kernel methods avoid the trainability issues of variational quantum eigensolvers and quantum neural networks, but face related expressibility-noise tradeoffs. Our kernel concentration result (Proposition II E) parallels barren plateau phenomena in variational circuits, suggesting that expressibility-related challenges may be fundamental to parameterized quantum circuits regardless of the training paradigm.

Several limitations of our study warrant mention. First, our theoretical separation results rely on specific circuit constructions that may not be optimal for practical problems. Second, our experiments used superconducting qubit platforms; trapped-ion or photonic systems may exhibit different noise characteristics affecting our conclusions. Third, we focused on binary classification; extension to multiclass problems and regression tasks requires additional investigation.

Future directions include:

- (i) Development of problem-specific quantum feature maps that maximize the alignment between task structure and quantum kernel geometry while minimizing required circuit depth.
- (ii) Investigation of fault-tolerant quantum kernel methods, where error correction overhead may be offset by the fixed-circuit nature of kernel evaluation.
- (iii) Theoretical characterization of the classical hardness of specific quantum kernel computations, moving beyond cryptographic assumptions to more practical complexity-theoretic foundations.
- (iv) Hybrid classical-quantum kernel approaches that use shallow quantum circuits for feature extraction combined with classical kernel post-processing.

In conclusion, quantum kernel methods represent a promising but challenging path toward practical quantum machine learning advantage. Our work provides both theoretical foundations and experimental benchmarks that can guide the development of quantum hardware and algorithmic approaches toward realizing this potential.

IV. METHODS

A. Proof of Theorem II A

We establish the exponential separation in effective dimension through a construction based on instantaneous quantum polynomial (IQP) circuits [23].

Consider a data encoding circuit of the form:

$$U(\mathbf{x}) = H^{\otimes n} \left(\prod_{i < j} e^{i\theta_{ij}(\mathbf{x})Z_iZ_j} \right) H^{\otimes n}, \quad (11)$$

where $\theta_{ij}(\mathbf{x}) = x_i x_j$ for input features $\mathbf{x} = (x_1, \dots, x_n)$.

The resulting quantum kernel takes the form:

$$\kappa_Q(\mathbf{x}, \mathbf{x}') = \left| \frac{1}{2^n} \sum_{\mathbf{z} \in \{0,1\}^n} e^{i \sum_{i < j} (\theta_{ij}(\mathbf{x}) - \theta_{ij}(\mathbf{x}')) z_i z_j} \right|^2. \quad (12)$$

For uniformly distributed inputs $\mathbf{x} \sim \mathcal{U}([0, 2\pi]^n)$, the kernel matrix eigenvalues $\{\lambda_k\}$ can be analyzed via random matrix theory. The key observation is that the phase structure $\sum_{i < j} (\theta_{ij}(\mathbf{x}) - \theta_{ij}(\mathbf{x}')) z_i z_j$ generates $2^{n(n-1)/2}$ distinct Fourier modes, leading to an eigenvalue distribution with support extending to exponentially small values.

Specifically, we show that the number of eigenvalues exceeding $\lambda > 2^{-n/2}$ scales as $\Omega(2^{n/2})$. The effective dimension at regularization $\lambda = 2^{-n/2}$ then satisfies:

$$d_{\text{eff}}(\lambda) = \sum_k \frac{\lambda_k}{\lambda_k + \lambda} \geq \sum_{k: \lambda_k > \lambda} \frac{1}{2} = \Omega(2^{n/2}). \quad (13)$$

For classical kernels computable in polynomial time, the associated feature maps have dimension at most $\text{poly}(n)$, bounding the rank and hence effective dimension of the kernel matrix.

The full proof requires careful analysis of eigenvalue concentration and appears in the Supplementary Information.

B. Proof of Theorem II C

Under a local depolarizing noise model, each two-qubit gate G is replaced by:

$$\mathcal{E}_p(G\rho G^\dagger) = (1-p)G\rho G^\dagger + \frac{p}{15} \sum_{P \in \mathcal{P}_2 \setminus \{I\}} P G \rho G^\dagger P, \quad (14)$$

where $\mathcal{P}_2$ denotes the two-qubit Pauli group.

For a circuit with G sequential two-qubit gates, the output state fidelity with the ideal state satisfies:

$$F = \langle \psi_{\text{ideal}} | \rho_{\text{noisy}} | \psi_{\text{ideal}} \rangle \geq (1-p)^G. \quad (15)$$

The quantum kernel involves two circuit executions (for $U(\mathbf{x})$ and $U(\mathbf{x}')$), giving total gate count $2G$. The kernel value can be written as an inner product between (potentially mixed) output states, and the distortion follows from the fidelity bound:

$$|\kappa_Q^{\text{ideal}} - \kappa_Q^{\text{noisy}}| \leq 2\sqrt{1 - F^2} \quad (16)$$

$$\leq 2\sqrt{1 - (1-p)^{4G}} \quad (17)$$

$$\leq 2\sqrt{4pG} \quad \text{for } pG \ll 1 \quad (18)$$

$$= 4\sqrt{pG}. \quad (19)$$

A tighter analysis using the contractivity of the depolarizing channel yields the stated bound $\Delta_\kappa \leq 1 - (1-p)^{2G}$.

C. Experimental Setup

1. Hardware Platform

Experiments were conducted on IBM Quantum Falcon-series processors accessed via the IBM Quantum Network. Device specifications at the time of experiments:

- Qubit count: <INSERT MEASURED VALUE> qubits
- Median T_1 : <INSERT MEASURED VALUE> μs
- Median T_2 : <INSERT MEASURED VALUE> μs
- Median single-qubit gate error: <INSERT MEASURED VALUE>
- Median two-qubit gate error: <INSERT MEASURED VALUE>
- Median readout error: <INSERT MEASURED VALUE>

Qubit selection for each experiment prioritized connected subgraphs with lowest average two-qubit gate error rates.

2. Circuit Implementation

Quantum kernels were implemented using PennyLane version <INSERT MEASURED VALUE> with the `qiskit.ibm` device backend. The following code snippet illustrates the kernel circuit construction:

```
import pennylane as qml
from pennylane import numpy as np

def quantum_kernel_circuit(x1, x2, n_qubits, n_layers):
    """Compute quantum kernel via swap test."""

    # Ancilla qubit for swap test
    qml.Hadamard(wires=0)

    # Encode x1 into register 1
    for layer in range(n_layers):
        for i in range(n_qubits):
            qml.RZ(x1[i % len(x1)], wires=i+1)
        for i in range(n_qubits - 1):
            qml.CNOT(wires=[i+1, i+2])

    # Encode x2 into register 2
    for layer in range(n_layers):
```

```

for i in range(n_qubits):
    qml.RZ(x2[i % len(x2)], wires=i+1+n_qubits)
for i in range(n_qubits - 1):
    qml.CNOT(wires=[i+1+n_qubits, i+2+n_qubits])

# Controlled swap
for i in range(n_qubits):
    qml.CSWAP(wires=[0, i+1, i+1+n_qubits])

qml.Hadamard(wires=0)
return qml.expval(qml.PauliZ(0))

```

3. Classical Baselines

Classical kernel methods were implemented using scikit-learn version 1.2.0. Hyperparameters were selected via 5-fold cross-validation with the following search grids:

- RBF kernel: $\gamma \in \{10^{-4}, 10^{-3}, \dots, 10^2\}$
- Polynomial kernel: $\text{degree} \in \{2, 3, 4, 5\}$, $c \in \{0, 1, 10\}$
- SVM regularization: $C \in \{10^{-2}, 10^{-1}, \dots, 10^4\}$

D. Dataset Generation

1. Synthetic Parity Dataset

Data points $\mathbf{x} \in \{-1, +1\}^d$ were generated uniformly at random. Labels were assigned according to:

$$y = \prod_{i \in S} x_i, \quad (20)$$

where $S \subset \{1, \dots, d\}$ is a hidden subset of size $|S| = k = \langle \text{INSERT MEASURED VALUE} \rangle$. The subset S remained fixed across training and test sets.

2. Discretized Continuous Symmetry Dataset

Data points were generated as rotated versions of a reference pattern $\mathbf{r} \in \mathbb{R}^d$:

$$\mathbf{x}_\theta = R(\theta)\mathbf{r} + \boldsymbol{\epsilon}, \quad (21)$$

where $R(\theta)$ applies rotation by angle θ in a randomly chosen 2D subspace and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2 I)$ adds noise. Labels were assigned as $y = \text{sign}(\cos(2\theta))$, creating two classes corresponding to angular sectors.

3. Molecular Property Dataset

Molecular fingerprints were computed using the RDChEMBL library for a curated subset of the ChEMBL database [24]. Binary labels indicated binding affinity above/below a threshold value for a selected protein target. Feature vectors of dimension $d = \langle \text{INSERT MEASURED VALUE} \rangle$ were generated using extended-connectivity fingerprints (ECFP4).

E. Statistical Analysis

All reported accuracies represent means over $\langle \text{INSERT MEASURED VALUE} \rangle$ independent train/test splits. Error bars indicate 95% confidence intervals computed via bootstrap resampling with $\langle \text{INSERT MEASURED VALUE} \rangle$ bootstrap samples. Statistical significance of quantum versus classical performance differences was assessed via paired t -tests with Bonferroni correction for multiple comparisons.

Kernel distortion measurements were averaged over $\langle \text{INSERT MEASURED VALUE} \rangle$ randomly sampled pairs $(\mathbf{x}, \mathbf{x}')$ from each dataset. Hardware kernel values were computed from $N_{\text{shots}} = \langle \text{INSERT MEASURED VALUE} \rangle$ circuit executions per kernel entry.

AUTHOR CONTRIBUTIONS

[Author One] conceived the project, developed the theoretical framework, and wrote the manuscript. [Author Two] implemented the quantum circuits and conducted hardware experiments. [Author Three] performed classical baseline comparisons and statistical analysis. All authors contributed to interpretation of results and manuscript revision.

COMPETING INTERESTS

The authors declare no competing interests.

CODE AVAILABILITY

All code for quantum kernel implementation, classical baselines, and data analysis is available at [INSERT REPOSITORY URL]. The implementation uses PennyLane version $\langle \text{INSERT MEASURED VALUE} \rangle$ and Qiskit version $\langle \text{INSERT MEASURED VALUE} \rangle$.

DATA AVAILABILITY

Synthetic datasets were generated using the procedures described in Methods with random seeds provided in the code repository. The molecular property dataset is derived from ChEMBL (version $\langle \text{INSERT MEASURED VALUE} \rangle$) and processed data files are available upon reasonable request from the corresponding author.

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